

Gwendolyn Noria

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Flight-critical Avionics | Real-time Simulations | High-performance Graphics

Education

The Ohio State University

B.S. Computer Science and Engineering

Columbus OH

Expected graduation May 2028

Delaware Area Career Center — Cybersecurity Program

Aug 2021 - May 2023

Activities

Avionics Lead Software Engineer, [Buckeye Space Launch Initiative](#), [IREC](#)

December 2025 - Present

- Led avionics software development for a 45,000 ft multistage competition rocket within a multidisciplinary engineering team.
- Responsible for end-to-end avionics firmware including control systems, telemetry, and data pipelines
- Architected our firmware to leverage parallelism enabling 500hz data acquisition, 100hz control loops.
- Verified and Validated Air brake Control Systems through MIL, SIL, PIL, and HIL industry testing.
- Engineered data ingestion pipeline to enable full post-flight state reconstruction upon rocket de-integration.
- Designed fault-tolerant logging system to capture and diagnose in-flight failures in firmware.
- Technologies used: C/C++, Zig, FreeRTOS, and Arduino.

Projects

[Ceresvoxel](#), Personal Project

Feb 2025 - Present

- A video game computing real-time simulation of vehicle dynamics and factory manufacturing pipelines.
- Analytical Orbital Mechanics using Patched Conics.
- No memory allocations occur during game runtime.
- Optimized rendering by using push constants for chunk indexing on the GPU.
- All static geometry is rendered using multidraw calls.
- Technologies used: Zig, Jolt, Vulkan, and RenderDoc.

[Mitre Embedded Capture the Flag](#), The Embedded Security Club at tOSU

Jan 2022 - May 2024

- Placed 9th among 80 teams in 2023.
- Defended against Carnegie Mellon's PPP without private key compromise during competition.
- Implemented key exchanges and measures to mitigate hardware access of critical systems

Work Experience

Peak Surfer Internship, National Security Agency

March 2026 - Present

- Provided specifications for firmware deliverables and provided gold-standard references for ciphers.
- Mentored 4 students to implement cryptographic ciphers on FPGAs and Software Defined Radios.
- Technologies used: 32 bit KAJU cipher and 128bit SIMON cipher

Capstone Mentor, The Ohio State University ECE department

Jan 2026 - Present

- Mentored senior undergraduates in firmware and embedded systems design
- Facilitated meeting deliverable deadlines for several project teams over the course of 5 months

Student Asset Specialist, Office of Technology and Digital Innovation

Aug 2024 - Feb 2026

- Facilitated the distribution of hardware across the OSU campus on tight deadlines
- Planned logistical itineraries and coordinated with a large team to deliver products on a daily basis

Skills

Programming Languages: C/C++ (advanced), Zig (advanced), Java (intermediate), and Python (intermediate)

Graphics & Simulation: Vulkan (advanced), OpenGL (intermediate), and Jolt

Tools: Linux (advanced), QEMU, LibVirt, RenderDoc

Systems: RTOS (FreeRTOS), control systems, firmware architecture, game engine architecture